

Question	Answer	Marks
6(a)	joule/coulomb	B1
6(b)(i)	$7.0 = (I \times 5.2) + (I \times 6.0) + 1.4$	C1
	$I = 0.50 \text{ A}$	A1
6(b)(ii)	$R = 1.4/0.50$ $= 2.8 \Omega$	A1
6(b)(iii)	$P = EI$ or $P = VI$ or $P = I^2R$ or $P = V^2/R$	C1
	efficiency = $[(0.50^2 \times 6.0)/(7.0 \times 0.50)] (\times 100)$ or efficiency = $[(0.50 \times 3.0)/(7.0 \times 0.50)] (\times 100)$ or efficiency = $[(3.0^2/6.0)/(7.0 \times 0.50)] (\times 100)$	C1
	efficiency = 43%	A1
6(b)(iv)	$R = \rho l/A$	C1
	$\alpha = \rho/R$ $= 3.7 \times 10^{-7}/6.0$ $= 6.2 \times 10^{-8} \text{ m}$	A1

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Question	Answer	Marks
3(c)	4: (cross-sectional) area (of wire)	B1